



Addressing UK Electricity's Price-Amplification

Analysis by the **Humane Energy** Think-Tank

The Humane Energy Project is part of the [Humane Economics](#) network



Addressing UK Electricity's Price-Amplification:

Additional Measures to Stabilise & Reduce Electricity Price Rises by up to 33%, without additional costs or Tax Sacrifices

Unprecedented prices for gas have brought unforeseen consequences, significantly raising the prices paid for electricity produced without gas. Applifying the effect of the gas price increases.

Our high level analysis, harnessing data captured by MyGridGB.co.uk since 2015, has identified the means to tackle this and significantly reduce the planned 1st October rises in energy bills, thus reducing the rise in electricity unit costs by up to 33% (which would equate to an 18% reduction in the price cap).

The measures proposed here could:

- **Cut the October electricity price rise by up to 33% (from 52p to 35p per unit) through the winter. Also reducing the April 2023 rise by up to 40%.**
- **Make a significant dent in the overall Energy Cap, from October 1st by up to an estimated £660 by October, and potentially £1,200 by April.**
- **Achieve more than double the savings that can be delivered by removing the green levy, and quadruple the savings from removing VAT (proposed by Conservative Leadership candidates Liz Truss and Rishi Sunak) which can be implemented in *addition* to those measures, *because*:**
 - **Zero cost: Funded by a reduction in abnormally high payments to non-gas producers. So without cost to the public purse or sacrifice of tax revenues, enabling this to be an additional measure *benefiting homes and businesses as well as the economy as a whole***
- **Further and significant reductions would be possible if the price of domestic gas sold into the domestic market was also capped.**
- **Provide a valuable buffer mitigating the effects of further supply shocks such as prolonged outages on the Nord Stream 1 pipeline from Russia to Europe.**



The team, led by Humane Energy think-tank founder Barry E James, chair of [BBFTA.org](https://bbfta.org), was supported by a team of experts including analysis by global energy expert and founder of [MyGridGB.co.uk](https://mygridgb.co.uk) Dr Andrew Crossland who has long campaigned, highlighting the risks of gas controlling the UK electricity market, and Andy Caulton who is the Founder & CEO of Hope Energy.

The proposal decouples the wholesale price of electricity from gas prices. Doing so could significantly **reduce the burden of shock energy price rises on households and businesses, significantly more than can be achieved by combining the measures proposed by the Conservative leadership candidates.**

Unlike measures proposed by the candidates, our proposal can be implemented **without forgoing tax revenues or adding costs to the public purse**. This is because they tackle the amplification effect on electricity prices of the current model. It achieves genuine over-pricing reductions, properly moderating the overheated market. They could therefore be implemented in addition to other such measures.

Decoupling from Gas Prices Fixes the UK Electricity's Price-Amplification Problem

We propose the means to decouple the wholesale cost of electricity from other sources from the price of gas. This can be via temporary price caps or harnessing already established CfD (Contract for Differences) mechanisms already in use elsewhere. This addresses the 'price-amplification' problem for electricity, at source and without having to continue to fund it from either from consumer's pockets, businesses or taxpayers.

How UK Wholesale Electricity Pricing Works

The price of energy is not determined in a traditional manner. The electricity market is determined on **a marginal clearing market**. The most expensive power station sets the price for the rest of the market. For example if three wind farms bid into the market at £10, a nuclear plant at £20 and a gas plant at £3,000, then all power stations get paid £3,000 if gas is dispatched. While effective in normal conditions, encouraging the cheapest and most reliable generators onto the system, **it means that in abnormal times, abnormal prices impact the whole market.**

A fuller [explanation is available here.](#)



Proposed Measures

Our proposal is for measures to decouple the price of electricity produced from gas from that produced from other sources (see below), including renewable and low carbon sources, which comprise around 50% of electricity:

There are a number of existing mechanisms which could be used right now to do so, including but not limited to:

1. 6-month duration contracts for difference for nuclear, wind, solar, biomass, hydro and storage.
2. A cleared price cap for nuclear, wind, solar, biomass, hydro and storage.

These measures could be implemented during September 2022 and then also form the basis for, or part of, a wider long term review by Ofgem, who's [current consultation](#) says, at Section 3.26 page 22, "*Ofgem will continue to evaluate reform options, including consideration of responses to the Call for Input, and further stakeholder engagement, with a view to arriving at conclusions by early 2023.*"

Similar measures are now being considered by the EU¹.

¹ <https://www.ft.com/content/02f848fc-3b80-4ddc-ba4f-26109d79db89>



Calculation Basis and Impact Range

Our analysis, based on data from MyGridGB, looks at how much electricity prices can be reduced if all electricity sources (except gas, coal and imports) are temporarily capped to a lower rate, so they're unaffected by the gas price.

Scenarios from our analysis include:

- **Aggressive:** Decoupling by **capping to £50/MWh** would reduce the electricity unit prices by almost **33% from October 1st, and around 40% from April 1st 2023.**
- **Moderate:** Decoupling to more normal prices (e.g. £100/MWh for wind and solar to £100/MWh and £50/MWh for the rest of the market) would reduce the electricity unit prices by almost **23% from October 1st, and around 33% from April 1st 2023.**
- **Conservative:** Decoupling by **capping to £100/MWh** (a typical renewable contract) for non-gas generators would reduce the unit cap price by almost **15% from October 1st, and around 27% from April 1st 2023.**

UK Households

Given that a typical domestic household in the UK consumes **3,100kWh** of electricity per year (according to [Energy UK](#)), this gives an **average saving per household of up to £527 from 1 October and £953 from 1 April 2021.**

Nationally

Given that (according to the [Office for National Statistics](#)) there were 28.1m UK households in 2021, and rising, **this could represent savings of up to £15B / year for households from October and up to £27B from April 2023.**

Businesses

This is hard to estimate given the commercially confidential nature of arrangements but the impact for SMEs is likely to be greatest (as they have less negotiating power) and they may benefit even more than households.



Economic Impact

Ofgem have said that the price cap will increase by a further 80% this October which will cause severe hardship for lower income families, particularly in northern, rural and inner city areas. The price of energy impacts the entire economy and, unchecked, could cause inflation across the economy to hit extreme levels.

Further Measures

The measures we are proposing lay ground work for future long term policy measures such as:

- Protecting industry through limiting wholesale prices as industry is not protected by a price cap.
- Showing a basis for how gas produced in the UK can be sold into the UK market at a different price to that sold internationally.
- To inform and accelerate the ongoing review by Ofgem of how the electricity market price is set by gas.²

We recommend urgent consideration of these measures with a view to early implementation.

Contact Information

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² www.ofgem.gov.uk/sites/default/files/2022-07/Net%20Zero%20Britain%20Publication%202022%20FINAL.pdf



About the Team



[Barry E James](#) Founder and Industrial Fellow at the University of East London's Centre of Fintech and Fellow at the Radix Think Tank. Barry has decades experience in technology, as a systems designer and entrepreneur, and in the support of innovators and entrepreneurs.

A technologist and changemaker for over three decades in 2012 he designed and proposed a major structural change to the UK Financial Regulator. This resulted in the creation of the FCA's Innovation Unit, as proposed, which has since been copied around the world, supporting financial innovation and Fintech.

In 2013 he helped cofound the Westminster Crowdfunding Forum which he co-chairs with Barry Sheerman MP, and provided the secretariat to the Associated APPG. He is the founding chair of [BBFTA.org](#) (The British Blockchain & Frontier Technologies Association) and founder from 2020 of both [Humane Economics](#) and now Humane Energy, both supported by BBFTA.

He has assembled a team of leading experts. including:

MyGridGB

Dr Andrew Crossland is a low carbon energy specialist who has developed low carbon energy projects across four continents. He has a PhD in how to use batteries to support the grid and has worked for companies in the UK and New Zealand. He formerly Directed Advance Further Energy Ltd who developed a UK wide calculator which underpins all domestic solar and battery sales. He also authored the book "Decarbonising Electricity Made Simple" which outlines how to get net zero electricity. Andrew is a Fellow of the Energy Institute, a Chartered Engineer and Associate Fellow of the Durham University Energy Institute. He also advises an energy conservation charity in Kenya.



Andy Caulton is Founder & CEO of Hope Energy, which plans to be the greenest energy supplier in the UK (and the world). Andy has been involved in the energy industry for 20 years, both within energy retail directly and as a consultant within a Big 4 consulting practice.



What are Humane Energy & Humane Economics? *(Thinking Deeper, for People & Planet)*



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A 'Think Tank'

A 'Do It Tank'

A Catalyst

An Incubator & Mentor

*For the transformation of energy from generation to capture & storage in
the UK and worldwide.*

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